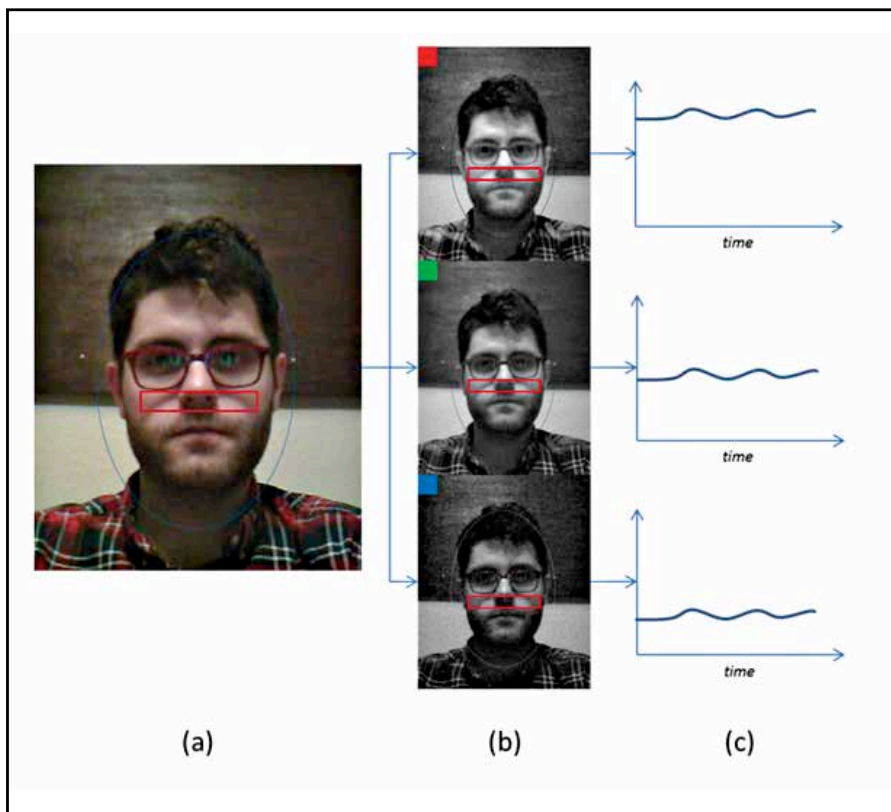
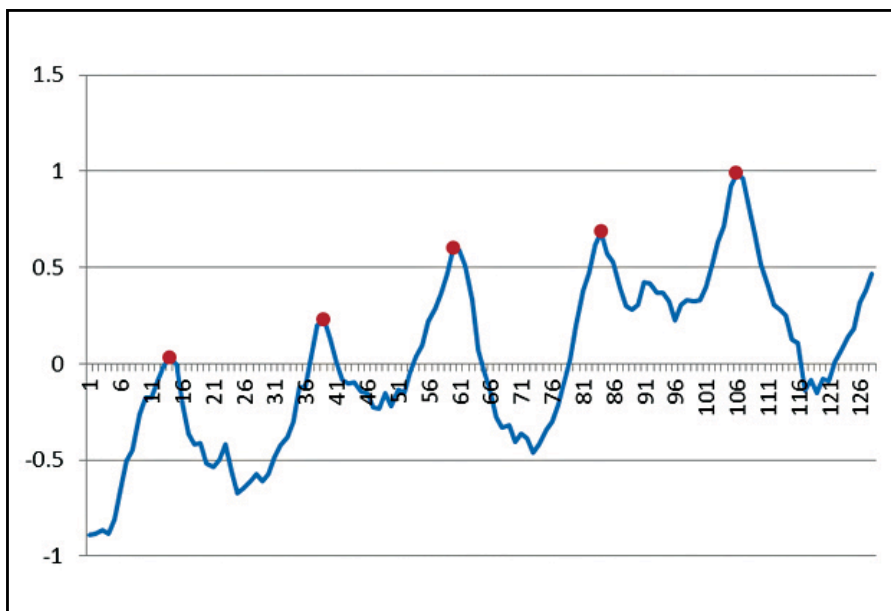




First stage of the pulse estimator pipeline to build waveforms after image analysis



Second stage of the pulse estimator with the processed waveform and identified features

like games and entertainment. The Perceptual Computing SDK is expected to eventually evolve into the RealSense™ SDK.

Some of Intel® RealSense™ Technology's highlights include an integrated 3D camera that has a best-in-class depth sensor, allowing new

ways to interact in gaming, entertainment, and content creation. Additionally, it allows finger tracking and recognizes facial expressions among numerous other features. This may seem trivial when you imagine it, but it's actually more than just gesture recognition. The

Nuance Dragon Assistant provides a voice-based natural language processing interface so that users can seamlessly interact using their voice. This allows a personalized conversational flow for interacting with the application, and can be integrated using the Intel RealSense™ technology on Intel-based devices. This year, there have also been a variety of partnership announcements with Microsoft for Skype video calling and collaborations with organizations such as Scholastic for developing educational software in addition to others such as AutoDesk, DreamWorks, Tencent, meta etc.

WHAT'S NEW?

Intel has been working on the Perceptual Computing SDK since a long time. It started as a library for pattern detection and recognition algorithms. With the development and release of the new RealSense™ hardware, there are some new additions to the Perceptual Computing SDK. These new additions allow even more flexibility and data to be available for further integration with your computing devices.

The new Pulse estimator module, which allows you to measure and determine the user's heart rates intuitively, is the newest feature in the camera. Although pulse estimation with excess motion is an issue, it can still measure the pulse rate seamlessly if the right settings are used, and that too without too many contraptions. Pulse estimation module is basically composed of a two-stage pipeline where the first stage receives input image sequences from the camera and in combination with the face tracking algorithm narrows down the face as a region of interest and further narrows down a part of the face as a sub-range. The sampling functions then divide the incoming image into individual color channels and perform a spatial average of all the pixels that fall within the dimensions of the sub-range. The sampler arrives at a single value for each color channel of the original frame, and logs these values over the course of many frames to build a "waveform" for each color channel. The second stage of the pipeline involves processing these samples to identify features of the waveform and thus estimate the pulse rate. For a detailed overview of all the functions and data structures that allow access to this data, you can visit https://software.intel.com/sites/landingpage/perceptual_computing/documentation/html/ for the SDK documentation.

The emotion detection module is another addition. It provides additional feedback by